The component model · branch · claude/design-system-foundation-HJb5Z

Lattice Design System

Function · Form · Substance · Finish. Authored in short names, organized in four layers.

What this deck covers.

01 The four layers — Function · Form · Substance · Finish

02 The seven function families

03 Components stay short — why dotted names lost

04 Manifests as the single source of truth

05 Discovery via the scaffolder and VS Code snippets

06 What ships, what's deferred

Section 01

The four layers

Each slide is a stack of four orthogonal decisions.

Function.

What the slide does for the audience. Seven families: Anchor, Statement, Inventory, Comparison, Progression, Evidence, Imagery.

Form.

The spatial composition that holds the content. Ten shapes: bookend, divider, canvas, grid, stack, ledger, panel, matrix, scatter, timeline, split.

Substance.

What fills the form. Four sources: prose, structure, series, graph — the engine's only plugin point.

Finish.

Palette tokens plus cross-cutting modifiers — dark, compact, mirror, accent, numbered, background flourishes.

Section 02

The seven functions

Pick by intent. Then by data shape.

01	**Anchor**	Orientation — title, divider, closing.
02	**Statement**	One claim — big-number, quote, split-panel.
03	**Inventory**	Parallel items — cards-grid, list-tabular.
04	**Comparison**	Contrast — compare-prose, verdict-grid.
05	**Progression**	Ordered movement — list-steps, roadmap.
06	**Evidence**	Data into picture — stats, kpi, quadrant, radar.
07	**Imagery**	Visual that carries meaning — image.

Section 03

Components stay short

Why dotted names lost the rethink.

DOTTED (REJECTED).

`<!-- _class: inventory.grid.cards compact -->` is 35 characters. Authors learn ~70 path combinations instead of 35 names. Implementation surfaces leak into invocation — `evidence.canvas.chart.progress` for what used to be `progress`.

SHORT (KEPT).

`<!-- _class: cards-grid compact -->` is 32 characters and what authors already know. Same pattern as shadcn, Chakra, Mantine, Lit, Slidev, Notion blocks. Discovery happens via tooling, not in the directive name.

Section 04

Manifests

Each layout ships a manifest at `lib/components/<bucket>/<name>/<name>.manifest.json`.

Single source of truth.

name, function, form, substance, purpose, variants, slots, skeleton. The pipeline is unchanged — manifests are metadata.

Consumed everywhere.

Scaffolder reads `skeleton`. Snippets read `skeleton` + `name`. The catalog groups by `function`. Editor autocomplete reads `variants`. Docs link from `docs`.

53 components shipped.

Every component in the galleries has one; a validation gate keeps them well-formed.

Section 05

Discovery

Two paths from blank deck to a slide skeleton.

CLI scaffolder

`npm run new:slide -- --list` prints the catalog grouped by function. `npm run new:slide -- cards-grid` emits the skeleton. Exit codes: 0 success, 1 unknown, 2 usage.

VS Code snippets

Type `lattice-` in any .md file. Autocomplete shows all 53 components by name and description. Tab inserts the canonical skeleton. Generated from manifests; freshness enforced by `npm run snippets:check`.

Section 06

Status

What ships on this branch.

- ✅ Spec — `design/design-system.md` with the four-layer model and component contract.
- ✅ Manifests — 53 components in `lib/components/<bucket>/<name>/<name>.manifest.json`, schema validator, loader.
- ✅ Scaffolder — `tools/new-slide.js` with `--list` and component-name modes.
- ✅ VS Code snippets — `.vscode/lattice.code-snippets` generated and committed.
- ✅ CLAUDE.md pointer — design-system.md at the top of "Read these first".
- ⛔ Templates.md reorganization by function family — deferred to a follow-up.
- ○ Visual catalog page (`docs/catalog.html`) — deferred to a follow-up.

design/design-system.md

Authors keep their short names. Everyone else gets a map.